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Tumour-proliferative fraction and growth factor expression as markers of tumour response to radiotherapy in cancer of the uterine cervix

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Abstract This study seeks to define the role of pre-treatment of evaluation of tumour growth fraction in cervical cancer and its relationship to the clinical course of the disease. In addition, it also seeks to explain whether cell kinetics and growth factor expression have an association with tumour response to radiotherapy and hence could be of value in the management of patients. All pre-treatment biopsies were analysed for the tumour-proliferative compartment by evaluation of Ki67 antigen expression and argyrophilic nucleolar organiser region (AgNOR) counts. Growth factor analysis was done by analysing for expression of epidermal growth factor (EGF), epidermal growth factor receptor (EGF-R) and transforming growth factors α and β (TGFα, TGFβ). A total of 152 patients were evaluated and a correlation obtained between pre-treatment status of the tumour-growth-fraction-associated markers and clinical outcome following radiotherapy. Such patients were either disease-free (group 1, $n = 106$) or with residual/recurrent disease (group 2, $n = 46$) at a 16-month follow-up. Pre-treatment analysis of AgNOR significantly correlated to disease status after treatment ($r = -0.517$, $P = 0.0000$). This may be due to an effect of cell proliferation. Lower AgNOR counts were significantly associated with recurrent/residual tumours, suggesting that increased proliferative activity may be a positive prognostic indicator. Similar results were also obtained for the other proliferation-associated marker Ki67 ($r = -0.443$, $P = 0.0000$). Expression of EGF and EGF-R also showed significant pre-treatment correlations with the final disease outcome ($r = 0.248$, $P = 0.031$ and $r = 0.503$, $P = 0.0000$ respectively). Both these markers were expressed more by patients belonging to group 2. The opposite was the case for TGFα, where patients belonging to group 1 showed higher values ($r = 0.417$, $P = 0.0001$). The other growth factor investigated, TGFβ, also showed a conspicuous differential expression in the two groups of patients ($r = -0.604$, $P = 0.0000$). Group 1 patients showed mostly mild to moderate expression while most group 2 patients were negative for the growth factor. It therefore appears that tumours with high AgNOR counts and Ki67 index, along with expression of the two types of transforming growth factor (α and β), responded better to radiotherapy.

Key words Cervix · Proliferation · Growth factor · Radiotherapy · Prognosis

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Introduction

The variation in tumour response to radiotherapy, both within and between tumour types is clinically evident. It is well recognized that factors such as histological type, stage, and degree of differentiation, tumour volume and growth pattern, among others, account for the reponse of human tumours to radiation (Davidson et al. 1992; Prempree et al. 1983). These clinically and paraclinically measurable criteria have been identified from retrospective studies, on the basis of enormous clinical radiotherapeutic experience gathered over many decades, and have served for qualitative classification of tumours as a function of their radiotherapeutic response. Although that knowledge is basically empirical, at present it plays a major role in the choice of cancer treatment modality. However, through application of this routine clinical approach in cancer radiotherapy, it has been clearly recorded that considerable heterogeneity in the response to radiation exists even within tumours, representing the same histological, anatomical and clinical features.

The basis for the variations in radio-response of individual tumours within a specified group is not completely understood, but it is certainly one of the most important factors in the success or failure of cancer radiotherapy. Therefore, from a clinical viewpoint, the ability to predict the response of an individual tumour to radiotherapy from one or more characteristics would be an obvious clinical advantage, because tumours expected to show higher radiosensitivity can be given priority in the schedule if there has to be a period of waiting (which is often the case in radiotherapy departments not only in developing countries but also in many developed countries), and patients identified as unlikely to be cured (or benefit) by conventional radiotherapy may be considered for other adjuvant modes of therapy or more aggressive radiotherapy protocols.

The total proliferative compartment of the tumour is dependent on a number of factors. In the present study, AgNOR counts and Ki67 antigen expression were used as markers of the total proliferative compartment. Ki67 is a proliferation-associated antigen expressed by most proliferating cells and is reported to give an accurate picture of the tumour growth fraction (Gerdes et al. 1984). Argyrophilic nucleolar organiser regions have also been shown to correlate with the proliferative phase of cells and provide yet another parameter to assess the tumour growth fraction (Eagen and Crocker 1992). The silver reaction product has been found ultrastructurally to be located in distinct compact regions within nucleoli. We have previously shown the importance of such markers in the evaluation of tumour progression and how they directly reflect the proliferating compartment (Lakshmi et al. 1993, 1996; Kesari et al. 1997). Sporn and Roberts (1985) have postulated the ability of cancer cells to produce and respond to their own growth factors by autocrine secretion. It is known that many types of tumour cells release polypeptide growth factors into their surroundings and that the same tumour cells often possess functional receptors for the released factor. It is therefore apparent that the balance between the various growth factors and receptors will influence the biological characteristics of a tumour. Peptide growth factors and receptors that function in this sort of a autocrine mechanism include epidermal growth factor (EGF), epidermal growth factor receptor (EGF-R) and the transforming growth factors α and β (TGFα and TGFβ). This study seeks to define the role of the tumour growth fraction and growth factors in cervical cancer and its relationship to radiotherapy response.

Materials and methods

Patients

A total of 152 patients with stage IIB or III (FIGO) were evaluated in the study and pre-treatment results correlated to radiotherapy outcome. A punch biopsy was taken from all patients, fixed in buffered formalin and processed for paraffin-wax embedding. Sections 5 μm thick were cut from the paraffin-embedded tissue and one section was stained by routine haematoxylin/eosin staining for histopathological evaluation. Duplicate serial sections were used for immunocytochemistry. All patients in the study received radical radiotherapy (40 Gy in 20 fractions for external-beam radiotherapy and 33 Gy to point A by Selectron - ICR).

Immunocytochemical localization of Ki67, EGF, EGF-R, TGFα and TGFβ

Immunocytochemical analysis was carried out as described by us earlier (Kesari et al. 1997; Lakshmi et al. 1996). Briefly, sections were dewaxed in xylene and hydrated through graded alcohols to deionised water. Endogenous peroxidase was blocked by a 25-min incubation in 3% H_2O_2 in methanol. The sections were rinsed with distilled water and then incubated with 0.3% bovine serum albumin to reduce non-specific antibody binding. Sections were incubated overnight at 4°C with monoclonal antibodies (dilution 1:20) to the Ki67 antigen (Novocastra, New Castle, UK), EGF, EGF-R, TGFα, TGFβ (Oncogene Science, Uniondale, NY, USA). Sections were then incubated with biotinylated anti-(mouse lg) at a dilution of 1:200 and peroxidase-conjugated streptavidin at 1:500 (Dako Strept ABComplex HRP Duet; Dako A/S, Denmark) for 30 min each at room temperature. Washing was carried out in phosphate-buffered saline after each step and the peroxidase reaction was developed by application of diaminobenzidine solution (Vector Laboratories, Calif., USA). The reaction was allowed to develop for 20 min, after which it was stopped by washing in distilled water. The sections were then lightly counterstained with Mayer's haematoxylin, dehydrated in ascending grades of alcohol, fixed in xylene and mounted in Distrene Dibutyl phthalate Xylene. Grading of the immunoreactivity of growth factors was done as previously explained by us (Kesari et al. 1996). Briefly, samples with less than 10% positive cells were considered negative. Samples with 11%–30% were considered as showing mild expression, 30%–50% moderate expression and those with over 51% positive cells as showing intense expression.

AgNOR evaluation

Silver staining for the detection of AgNOR was carried out by the method of Croker et al. (1989) and elaborated by us earlier (Kesari et al. 1997; Lakshmi et al. 1993).

Data analysis

Statistical analysis by the Kruskal Wallis one-way analysis of variance and Spearman correlation was done to look for relationships between the markers studied and the clinical outcome of the patient. For assessing odds ratio, Fisher's exact test was used.

Results

The 152 patients evaluated for a correlation between pretreatment experimental analyses and the clinical status of the patient after radiotherapy fell into two categories: group 1 included 106 patients who were disease-free after radiotherapy (and during a follow-up of 16 months), and group 2 comprised 46 patients who either had residual disease or developed recurrent disease within 16 months of treatment.

Immunoreactivity clearly showed that Ki67 was nuclear (Fig. 1a). A labelling index (LI) based on the percentage of positive cells was calculated for Ki67. Depending on the extent of the LI, patients were divided into two groups: a low-LI group and a high-LI group, the cut-off value being the median obtained for the particular marker. For Ki67, in group 1 the median LI

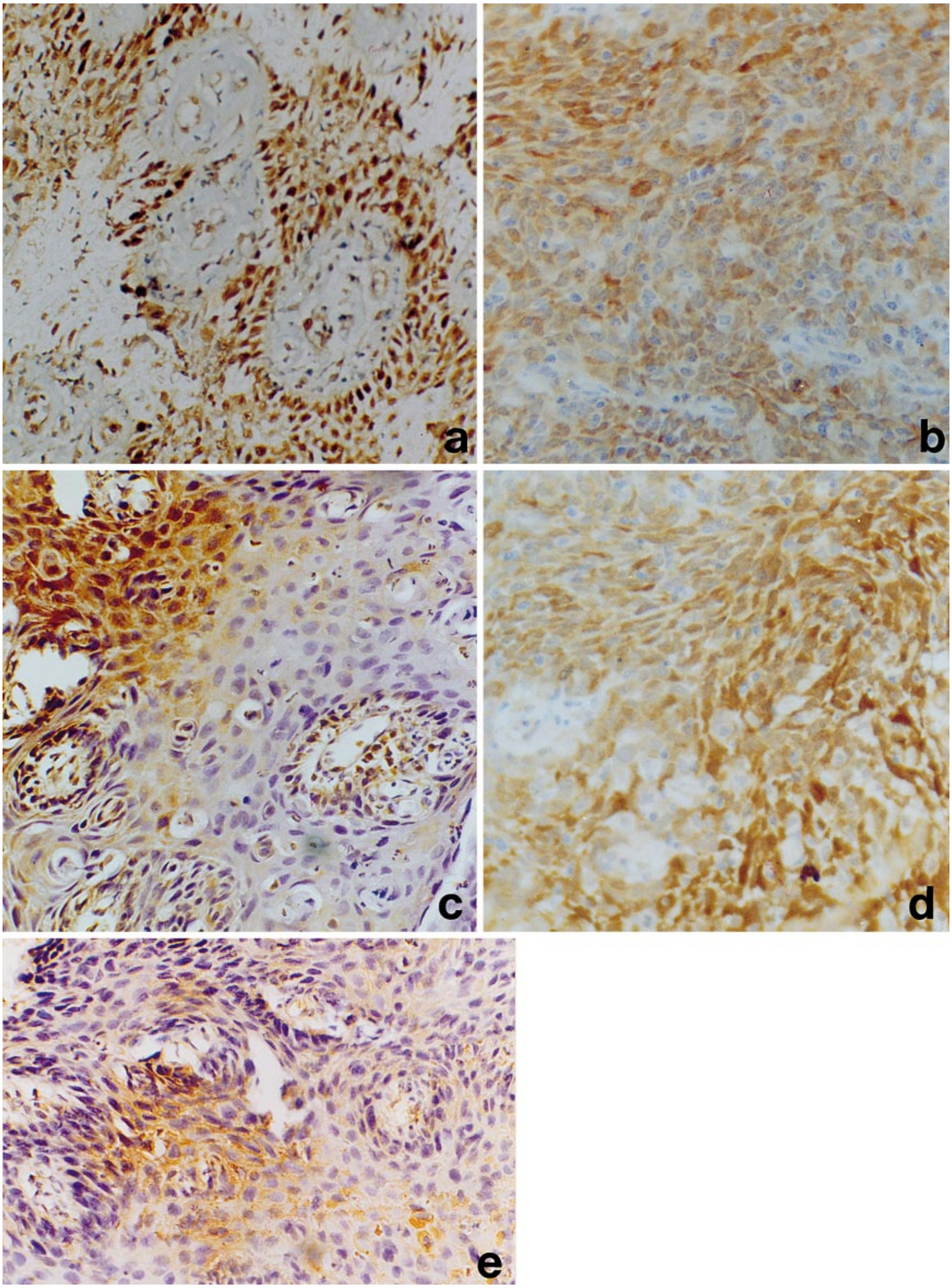

Fig. 1 **a** Nuclear immunoreactivity for Ki67 expression in invasive carcinoma of the uterine cervix (365×). **b** Intense cytoplasmic immunoreactivity of epidermal growth factor in invasive carcinoma of the uterine cervix (365×). **c** Intense cytoplasmic immunoreactivity of transforming growth factor α in invasive carcinoma of the uterine cervix (365×). **d** Intense membrane and cytoplasmic immunoreactivity of epidermal growth factor receptor in invasive carcinoma of the uterine cervix (365×). **e** Typical cytoplasmic immunoreactivity of transforming growth factor β in invasive carcinoma of the uterine cervix (365×)

value was 48, with 49% of patients in the lower LI category and 51% in the higher LI category. In group 2, the Ki67 median LI value was 36, with 65% of patients in the lower LI category and 35% in the higher LI category. These results, summarized in Table 1, showed a correlation to disease outcome following radiotherapy ($r = -0.443$, $P = 0.0000$). The other proliferation-related parameter, AgNOR counts, also showed signifi-

Table 1 Ki67 labelling index (*LI*) in cervical cancer patients with respect to disease status after radiotherapy. *Group 1* patients remaining disease-free after radiotherapy; *group 2* patients with residual/recurrent disease after radiotherapy

	Patient distribution (%)	
LI category	Group 1 (n = 106) median LI = 48	Group 2 (n = 106) median LI = 36
Lower	49	65
Higher	51	35

Table 2 Argyrophilic nucleolar organiser region (AgNOR) counts in cervical cancer patients with respect to diseasestatus after radiotherapy. *Group 1* patients remaining disease-free after radiotherapy; *group 2* patients with residual/recurrent disease after radiotherapy

	Patient distribution (%)	
AgNOR category	Group 1 (n = 106)	Group 2 (n = 46)
Lower	47.2	82.6
Higher	52.8	17.4

cant correlation to the disease status after radiotherapy ($r = -0.517$, $P = 0.0000$). As done for Ki67, the two groups were stratified into two categories, based on the median AgNOR count. For group 1 patients, the median AgNOR count was 3.3 while it was 2.7 for group 2. In group 1, 47.2% of patients were in the lower count category and 52.8% in the higher category. Similarly in group 2, 82.6% fell into the lower category and 17.4% into the higher category. These results are summarised in Table 2. To confirm these results further, the AgNOR counts were graded in an alternative way: on a continuous scale in three categories, as ≤2, 2.1–4 and >4.1. Here again, as evident from Table 3, patients in group 2 were significantly associated with lower AgNOR counts. To estimate the prognostic significance (risk involved) of these results, the data were analysed by Fisher's exact test. The odds ratio of a tumour with a low Ki67 LI, responding poorly to radiation therapy, was 6.5 (9.5% Cl = 3.04, 13.94). Similarly a tumour with low AgNOR counts had an odds ratio of 6.2 of responding poorly to radiotherapy (95% Cl = 2.81, 13.60).

Table 3 AgNOR counts (further stratification) in cervical cancer patients with respect to disease status after radiotherapy. *Group 1* patients remaining disease-free after radiotherapy; *group 2* patients with residual/recurrent disease after radiotherapy

	Patient distribution (%)	
AgNOR count stratification	Group I (n = 106)	Group II (n = 46)
≤ 2	7.5	30.4
2.1–4	71.7	56.6
4.1 and above	20.8	13.0

Immunocytochemical analysis for the expression of the growth factors EGF, TGFα, TGFβ and the receptor EGF-R was also carried out (Fig. 1b–e). In order to obtain a quantitative idea to the immunocytochemical reaction, a grading of 1–4, previously described by us (Kesari et al. 1997; Lakshmi et al. 1996), was utilized, where 1 signified an insignificant reaction, 2 signified mild immunoreactivity, 3 signified moderate immunoreactivity and 4 represented intense immunoreactivity.

Expression of growth factors in the two groups of patients also provided interesting results (Table 4). Analysis of EGF showed the factor to be significantly correlated to disease outcome ($r = 0.248$, $P = 0.031$).

Table 4 Comparison of growth factors in patients remaining disease-free and patients with residual/recurrent disease. Expression index: 1 negative, 2 mild, 3 moderate, 4 intense. *EGF* epidermal growth factor, *TGF* transforming growth factor, *EGF-R* EGF receptor

	Patient distribution (%)		
Parameter	Expression index	Group 1 (patients with no disease after radiotherapy (n = 106)	Group 2 (patients with recurrent disease) (n = 46)
EGF	1	0	0
	2	9.4	0
	3	64.1	52.2
	4	26.5	47.8
TGFα	1	0	4.3
	2	20.8	56.6
	3	67.9	39.1
	4	11.3	0
EGF-R	1	0	0
	2	28.3	4.3
	3	58.5	30.4
	4	13.2	65.3
TGFβ	1	3.8	65.2
	2	62.3	30.4
	3	32.0	4.4
	4	1.9	0

As seen in Table 4, intense expression of EGF was more frequent in patients belonging to group 2. This association was even more evident in the analysis of EGF-R ($r = 0.503$, $P = 0.0000$). While most patients belonging to group 1 showed mild to moderate expression of EGF-R, a majority of group 2 patients showed intense expression of the receptor. A noticeable difference was also evident in the expression of TGFα. Both EGF and TGFα share the same receptor (EGF-R). However, TGFα was expressed more frequently in patients belonging to group 1 than in group 2 patients. The expression of TGFα also therefore correlated with the status of disease following radiotherapy ($r = 0.417$, $P = 0.0001$). The other growth factor analysed was TGFβ. Here again there were significant differences between the two groups. The majority of group 2 patients did not express TGFβ. Thus marker also showed very significant correlation ($r = 0.604$, $P = 0.0000$) to the status of disease after radiotherapy.

Discussion

Carcinoma of the uterine cervix is an attractive model system for studying the clinical applicability of laboratory-based predictive assays on tumour response to radiotherapy. It is a disease that, in many centres including ours, is managed primarily by radical radiation therapy. There exists, however, a wide variation in the response of tumours to radiotherapy. The dose that may be given is based on a balance between achieving tumour control and keeping normal tissue morbidity acceptably low. If the response of tumours to radiotherapy could be predicted, then the more radio-resistant tumour types could be considered for other treatment modalities including surgery and possibly adjuvant chemotherapy or by increasing the radiation dose given. Conversely radiosensitive tumours could, in theory, be given lower doses allowing corresponding reduction in normal tissue damage. The clinical correlations emerging for tumour and normal tissue radiosensitivity and patient response to treatment are therefore a critical issue in radiation biology. If achievable, such findings could bring about greater individualization of patient treatment by radiotherapy.

The results presented so far have shown significant variations between the two groups of patients. Pretreatment analysis of AgNOR and Ki67 expression significantly correlated to disease status after treatment. This may be due to an effect of cell proliferation. Lower AgNOR and Ki67 counts were significantly associated with group-2 patients, suggesting that increased proliferative activity may be a positive prognostic indicator. These data suggest that tumours with high proliferative fractions are more radiosensitive. This could be due to the intrinsic radioresistance of low-growth-fraction tumours, which have more cells in the G0 phase. Such tumours have better DNA repair efficiency as well as larger hypoxic cell compartments (West 1995). Nakano and Oka (1993) have demonstrated that the Ki67 index positively correlates with prognosis and early response to radiation therapy. Contradictory results have been reported in lymphoma, breast cancer and soft-tissue sarcoma (Grogan et al. 1988; Ueda et al. 1989; Wintzer et al. 1991). Another study in oral squamous cell carcinoma by Valente et al. (1994) found the Ki67 index at diagnosis to have no significant influence on the tumour response to radiotherapy. However, there was a significant difference in the number of cells expressing Ki67 after a radiation dose of 10 Gy, which could significantly differentiate responders from non-responders.

It is not clear why differences are present in the proliferative fraction among tumours. The rates of cell proliferation are thought to be an inherent characteristic of a tumour that could be regulated both at the genetic level and by the tumour microenvironment or by a combination of the two. It is in this regard that the present data on the growth factors are interesting. Expression of EGF and EGF-R also showed significant pretreatment correlations with the final disease outcome. Both these markers were expressed more by patients who had residual/recurrent disease. The opposite was the case for TGFα, where patients who were disease-free showed higher values. Both EGF and TGFα share the common receptor EGF-R (Yoshida et al. 1990) and the reason for this differential expression needs to be investigated. It may be that this differential expression of EGF and TGFα have significant implications. The other growth factor investigated was TGFβ and its expression in the two group of patients was significantly different. Patients remaining disease-free showed mostly mild to moderate expression while most of the patients with residual/recurrent disease were negative for the growth factor. TGFβ is known to exert an antiproliferative effect and is known to promote differentiation (Comerci et al. 1996). We have previously shown expression of TGFβ to be important in the development of pancreatic cancer (Kesari et al. 1996). The reason for its differential expression in the present study among the two groups of patients is not clear. However, since its absence is associated with poor prognosis, TGFβ is a candidate marker for tumour response to radiotherapy.

The importance of knowledge of the radiation responsiveness status of human tumours as well as its role as a determinant for therapeutic modality planning has stimulated active research on predictive assays (West 1995). The main goal of this research is to develop laboratory tests, predictive of responsiveness to radiation therapy. This pilot study has shown the value of tumour cell proliferation and growth factor expression as markers for tumour response to radiotherapy in cancer of the uterine cervix. Further analysis of the details of why these markers are differentially expressed in different groups of patients is in progress. Such an approach will help improve the ability to predict treatment outcome and will allow early and better choice of the ideal treatment modality, thereby helping to improve the individualization of radiotherapy.

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